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Staff Software Engineer - LLM & Agentic AI

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COVER LETTER

Dear Hiring Manager,

I am a Staff Software Engineer with 13 years of experience in natural language processing and document intelligence, and I am writing to express my interest in senior and staff-level engineering roles in Applied AI, Machine Learning, and LLM and agentic systems. My focus throughout my career has been consistent: taking AI out of the research notebook and into production systems that run reliably, at scale, on real business data.

In my current role I lead LLM and multi-agent integration for a Document AI platform that classifies and extracts data across 500+ document types and 500+ data fields, processing millions of documents each week. I designed and deployed the production multi-agent orchestration layer on LangGraph, coordinating specialized classification, extraction, and validation agents through stateful graphs, LLM tool and function calling, and human-in-the-loop review. I built Model Context Protocol integrations so new platform capabilities are exposed to the models through a standard interface rather than bespoke glue code for each addition, and introduced context-aware guardrails and automated validation checks that sharply reduced hallucination rates on business-critical fields. Collectively this work raised end-to-end extraction accuracy by approximately 30%.

I regard the engineering beneath the models as equally important. The platform runs as an event-driven pipeline, moving documents from object storage through a queue into an asynchronous container-based worker pool that calls Gemini and Claude models on Amazon Bedrock within a private network. A shared resilience layer provides distributed rate limiting, exponential backoff with jitter, and a circuit breaker that applies backpressure to the queue rather than failing work downstream; prompts are versioned in a datastore, failed messages route to a dead-letter queue, and documents falling in the lower confidence band are routed to human review instead of being silently accepted. I am also well versed in the AWS agentic stack, including Amazon Bedrock AgentCore and AgentCore Runtime for managed agent hosting alongside the Strands Agents SDK.

Earlier, I designed and trained the custom LayoutLM, Donut, and UDOP document-understanding models behind an enterprise Document AI platform, improving extraction accuracy by 70% over the previous baseline, and subsequently made the case for replacing that custom-model approach with large language models as the technology matured. Before that I led knowledge graph and conversational AI engineering, building a Neo4j knowledge graph with custom named entity recognition, relation extraction, and entity linking, along with the intent-classification engine for an automated support assistant. I began my career in NLP and ML research at Tata Consultancy Services, where my work on mining process models from software test artifacts was published as four peer-reviewed IEEE papers. I hold Machine Learning (Silver Medal) and Deep Learning certifications from IIT Madras, and have received the Bravo Award three times and the XOR-Champ Award for outstanding work.

What I bring

- 13 years spanning classical machine learning, transformer-based document models, and production LLM and agentic systems.
- Production multi-agent orchestration on LangGraph: stateful graphs, tool and function calling, Model Context Protocol, agentic RAG, memory management, and human-in-the-loop review.
- AWS agentic stack and cloud delivery: Bedrock AgentCore, AgentCore Runtime, Strands Agents SDK, Amazon Bedrock, SageMaker, ECS, ECR, S3, and CloudWatch.
- Production reliability engineering: rate limiting, retries and backoff, circuit breakers, dead-letter queues, confidence-based routing, evaluation, and monitoring.
- Document AI and NLP depth: LayoutLM, Donut, UDOP, DocVQA, BERT, NER and entity linking, embeddings, vector search, and semantic similarity.
- Python, R, and Java; Docker, MLflow, and CI/CD; MySQL, MongoDB, Neo4j, and Elasticsearch.

I would welcome the opportunity to discuss how this experience maps to your team's needs. My resume, portfolio, and published work are available at the links above, and I am glad to provide further detail on any of the systems described here. Thank you for your time and consideration.

Sincerely,
Chetankumar Khadke